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pSIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Ganji Madhan kumar	**Reg. No.:**	192472328
Section / Batch:		**Date:**	09-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I ______________________ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ______________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

18-07-2026

Name: G. Madhankumar
Reg No: 192472328
Code: CSA0624
Subject: DAA

CO1 - AT1

- Application Based Questions

1. Online Video Streaming platform.

Given

$$T(n) = 2T(n/2) + n$$

Step:1 Identify Master Theorem parameters.

$a = 2$

$b = 2$

$f(n) = n$

Step:2 Compute

$$n^{\log_b a} = n^{\log_2 2} = n$$

$$f(n) = \Theta(n^{\log_b a})$$

This is Master Theorem Case 2.

Time Complexity

$$\boxed{T(n) = \Theta(n \log n)}$$

Scalability Analysis:

- performance grows almost linearly with an additional logarithmic factor.
- Suitable for handling millions of users efficiently.

Asymptotic Notation

$$\boxed{\Theta(n \log n)}$$

2. Social Media Feed Processing

Given:

$$T(n) = T(n-1) + n$$

Substitution Method

Expand recurrence:

$$T(n) = T(n-1) + n$$
$$= T(n-2) + (n-1) + n$$
$$= T(n-3) + (n-2) + (n-1) + n$$
$$\vdots$$
$$= T(1) + 2 + 3 + \cdots + n$$

Sum of first n numbers:

$$\frac{n(n+1)}{2}$$

Therefore,

$$\boxed{T(n) = \Theta(n^2)}$$

Performance Analysis:

- Running time increases quadratically.
- Large inputs significantly reduce performance.

Asymptotic Notation:

$$\boxed{\Theta(n^2)}$$

E-Commerce Recommendation System.

Algorithm 1

$\boxed{O(n \log n)}$

Algorithm 2

$\boxed{O(n^2)}$

Comparison

Algorithm	Growth Rate	Efficiency
$O(n \log n)$	Slow growth	Better
$O(n^2)$	Fast growth	poor

Analysis

- For small datasets, both perform similarly
- For large datasets, $O(n \log n)$ is much faster.

Conclusion

$\boxed{O(n \log n) \text{ is more scalable and efficient.}}$

4. Cloud Storage Processing System

Given

$$T(n) = 3T(n/2) + n$$

Parameters

$$a=3, b=2, f(n)=n$$

Compute

$$n^{\log_2 3} = n^{1.585}$$

$$n < n^{1.585}$$

$$f(n) = O(n^{1.585-\varepsilon})$$

This is Master Theorem Case 1.

Complexity $\boxed{T(n) = \Theta(n^{1.585})}$

Analysis:

- Better than Quadratic
- Suitable for large cloud storage systems.

5. Financial Analytics System

Given

$$T(n) = 2T(n/2) + n\log n$$

$$a=2,\ b=2,\ f(n) = n\log n$$

$$n^{\log_2 2} = n$$

$$f(n) = \Theta(n\log n) = \Theta(n^{\log_b a} \log^1 n)$$

This is Master Theorem Case 2 with $K=1$

Complexity

$$\boxed{T(n) = \Theta(n\log^2 n)}$$

Performance

- Efficient for large financial datasets
- Slightly slower than $n\log n$.

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6. Banking Transaction Processing System.

Given

$$T(n) = 4T(n/2) + n^2$$

$$a=4,\ b=2,\ f(n) = n^2$$

$$n^{\log_2 4} = n^2$$

$$f(n) = \Theta(n^{\log_b a})$$

Master Theorem Case 2.

Complexity

$$\boxed{T(n) = \Theta(n^2\log n)}$$

Analysis:

- Handles transactions reasonably well.
- Cost increases for extremely large transaction volumes.

Search Engine Index Builder

Given

$$T(n) = T(n/2) + 1$$

Substitution Method:

Expand:

$$T(n) = T(n/2) + 1$$
$$= T(n/4) + 2$$
$$= T(n/8) + 3$$
$$\vdots$$

After k expansions,

$$\frac{n}{2^k} = 1$$

Therefore,

$$k = \log_2 n$$

Hence,

$$T(n) = O(\log n)$$

Scalability

- Very efficient.
- performance increases slowly even for huge datasets.

Asymptotic Notation

$$\boxed{O(\log n)}$$

8. Gaming Engine Optimization

Algorithms:

- $O(n)$
- $O(n \log n)$
- $O(n^2)$

Comparison:

Complexity	Growth	Efficiency
$O(n)$	Linear	Best
$O(n \log n)$	Moderate	Good
$O(n^2)$	Quadratic	Worst

Analysis:

- High-resolution graphics require efficient algorithms.
- Linear algorithms scale best.

Conclusion:

$$\boxed{O(n) < O(n \log n) < O(n^2)}$$

Most efficient:

$$\boxed{O(n)}$$

Data Backup Optimization System.

Given

$$T(n) = 2T(n/2) + \sqrt{n}$$

$$a = 2, \; b = 2, \; f(n) = \sqrt{n} = n^{1/2}$$

Compute

$$n^{\log_2 2} = n$$

$$\sqrt{n} = O(n^{1-\epsilon})$$

Master Theorem Case 1.

Complexity

$$\boxed{T(n) = \Theta(n)}$$

Analysis:

- Nearly linear performance.
- Excellent scalability for large backups.

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10. Big Data Analytics Pipeline

Given

$$T(n) = T(n-1) + \log n$$

Substitution Method:

Expand: $T(n) = T(n-1) + \log n$

$$= T(n-2) + \log(n-1) + \log n$$

$$= T(n-3) + \log(n-2) + \log(n-1) + \log n$$

$$\vdots$$

$$= T(1) + \sum_{i=2}^{n} \log i$$

Using

$$\sum_{i=1}^{n} \log i = \log(n!) = \Theta(n \log n)$$

Therefore,

$$\boxed{T(n) = \Theta(n \log n)}$$

Scalability

- Efficient for big data
- Better than quadratic algorithms.

Asymptotic Notation:

$$\boxed{\Theta(n \log n)}$$